

Topology and Manifolds

MH3600

Chapter 3: Continuity and Metric Spaces

Assoc. Prof. François J. M. Gay-Balmaz

Division of Mathematical Sciences
School of Physical and Mathematical Sciences
Nanyang Technological University

Outline

- 1 Continuous functions
- 2 The metric topology
- 3 Continuity in metric spaces

Outline

- 1 Continuous functions
- 2 The metric topology
- 3 Continuity in metric spaces

Continuous functions

Continuity is a **fundamental concept in topology**. For example, when you hear that “a coffee cup and a doughnut are topologically equivalent”, this is really a claim about the existence of a certain continuous function. Recall that the standard definition of **continuity** of a **real valued function of a real variable** at a point $x_0 \in \mathbb{R}$ in the domain $\mathcal{D}(f) \subset \mathbb{R}$ of the function $f : \mathcal{D}(f) \subset \mathbb{R} \longrightarrow \mathbb{R}$, is as follows:

Definition

Suppose $f : \mathcal{D}(f) \subset \mathbb{R} \longrightarrow \mathbb{R}$ is a function and $x_0 \in \mathcal{D}(f)$. Then f is **continuous at point x_0** if for all $\varepsilon > 0$ there exists $\delta(\varepsilon) > 0$ such that for all $|x - x_0| < \delta(\varepsilon)$ and $x \in \mathcal{D}(f)$ we have $|f(x) - f(x_0)| < \varepsilon$.

We then say that f is **continuous at a set $A \subset \mathbb{R}$** if f is continuous at each point of A .

The following is a consequence of the previous definition:

Theorem

Let $f : \mathcal{D}(f) \subset \mathbb{R} \longrightarrow \mathbb{R}$. Then f is continuous on $\mathcal{D}(f)$ **if and only if** for all open sets $V \subset \mathbb{R}$, we have $f^{-1}(V)$ is open relative to $\mathcal{D}(f)$.

Inspired by this result, we have the following as our (standard) definition of continuous functions. We take functions defined on all of topological space X , so there is no need for the “open relative to” part of the above definition.

Definition

Let X and Y be topological spaces. A function $f : X \rightarrow Y$ is **continuous** if for each open subset V of Y , the set $f^{-1}(V)$ is open in X . The symbol $\mathcal{C}(X, Y)$ denotes the set of continuous functions between the topological spaces X and Y .

In Calculus 1, continuity is defined based on limits (which are defined using ε 's and δ 's), however we have not yet defined the limit of a function (though we have defined limit points of sets). Our definition is based entirely on open sets and we should in fact state that f is continuous **relative to the topologies on X and Y** .

Lemma

Let $f : X \rightarrow Y$, let $\mathcal{B}$ be a basis for the topology on Y , and let $\mathcal{S}$ be a subbasis for the topology on Y .

- (1)** *f is continuous if $f^{-1}(B)$ is open in X for each $B \in \mathcal{B}$.*
- (2)** *f is continuous if $f^{-1}(S)$ is open in X for each $S \in \mathcal{S}$.*

Proof of the lemma

(1) Let $V \subset Y$ be open. Then (by definition of basis) there are $B_\alpha \in \mathcal{B}$ for $\alpha \in J$ such that $V = \bigcup_{\alpha \in J} B_\alpha$. Then $f^{-1}(V) = f^{-1}(\bigcup_{\alpha \in J} B_\alpha) = \bigcup_{\alpha \in J} f^{-1}(B_\alpha)$ is an open set in X by hypothesis. So each $f^{-1}(B_\alpha)$ is open in X and $f^{-1}(V)$ is open in X . Hence f is continuous.

(2) Let $V \subset Y$ be open. Then (by the definition of subbasis) there are S_α^i for $\alpha \in J, i \in \mathbb{N}$ such that $V = \bigcup_{\alpha \in J} (S_\alpha^1 \cap S_\alpha^2 \cap \cdots \cap S_\alpha^{n_\alpha})$. Then

$$\begin{aligned} f^{-1}(V) &= f^{-1}\left(\bigcup_{\alpha \in J} (S_\alpha^1 \cap S_\alpha^2 \cap \cdots \cap S_\alpha^{n_\alpha})\right) = \bigcup_{\alpha \in J} f^{-1}(S_\alpha^1 \cap S_\alpha^2 \cap \cdots \cap S_\alpha^{n_\alpha}) \\ &= \bigcup_{\alpha \in J} (f^{-1}(S_\alpha^1) \cap f^{-1}(S_\alpha^2) \cap \cdots \cap f^{-1}(S_\alpha^{n_\alpha})) \end{aligned}$$

is open in X since each $f^{-1}(S_\alpha^i)$ is open in X by hypothesis and so $f^{-1}(S_\alpha^1) \cap f^{-1}(S_\alpha^2) \cap \cdots \cap f^{-1}(S_\alpha^{n_\alpha})$ is open for each $\alpha \in J$, and hence the union is open. So $f^{-1}(V)$ is open and f is continuous. ■

The new continuity definition and the old one

Example

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ (with $\mathbb{R}$ having the standard topology) be **continuous** according to our new definition. We now see that this definition implies the one based on ε - δ that you learnt in calculus.

Let $x_0 \in \mathbb{R}$ and $\varepsilon > 0$. Then the interval $V = (f(x_0) - \varepsilon, f(x_0) + \varepsilon)$ is open in the range space $\mathbb{R}$. Therefore, $f^{-1}(V)$ is open in the domain space $\mathbb{R}$. Now $x_0 \in f^{-1}(V)$ so there is some basis element (a, b) containing x_0 with $(a, b) \subset f^{-1}(V)$ (recall that the standard topology on $\mathbb{R}$ is defined as the topology with basis $\{(a, b) \mid a, b \in \mathbb{R}, a < b\}$). Choose $\delta = \min \{x_0 - a, b - x_0\}$. Then for $x \in (x_0 - \delta, x_0 + \delta)$ (that is, for $|x - x_0| < \delta$) we have $f(x) \in V$ (that is, $|f(x) - f(x_0)| < \varepsilon$).

So our definition of continuity implies the ε - δ definition of continuity. The converse implication is carried out in the tutorials.

Theorem

Let X and Y be topological spaces. let $f : X \rightarrow Y$. Then the following are equivalent:

- (1) f is continuous.
- (2) For every subset A of X , one has $f(\bar{A}) \subset \overline{f(A)}$.
- (3) For every closed subset B of Y , the set $f^{-1}(B)$ is closed in X .
- (4) For each $x \in X$ and each neighborhood V of $f(x)$, there is a neighborhood U of x such that $f(U) \subset V$.

Homeomorphisms

Definition

Let X and Y be topological spaces. Let $f : X \rightarrow Y$ be a bijection (one to one and onto). If both f and $f^{-1} : Y \rightarrow X$ are continuous, then f is a **homeomorphism**.

If $f : X \rightarrow Y$ is a homeomorphism then U is open in X if and only if $f(U)$ is open in Y . Any property in X that is expressed entirely in terms of the topology on X yields through the homeomorphism the corresponding property in Y . Such a property is called a **topological property** of X . Examples of such properties are **open/closed**, **limit points of a set**, **limits of a sequence**, **a basis or subbasis for the topology**, and as we will see later on, **connectedness and compactness**.

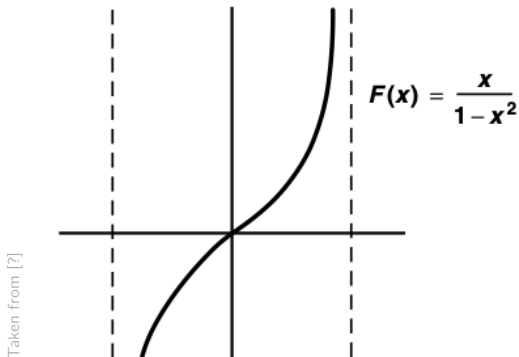
Topological embeddings

Definition

Let $f : X \rightarrow Y$ be an injective (one to one) continuous map. Let $Z = f(X)$ (so that f is onto Z) be considered a subspace of Y . Let $f' : X \rightarrow Z$ be the restriction of f to Z (so f' is a bijection). If f' is a homeomorphism of X with Z , then $f : X \rightarrow Y$ is a **topological embedding** (or simply **embedding**) of X in Y .

Example

Consider $F : (-1, 1) \rightarrow \mathbb{R}$ defined by $F(x) = x / (1 - x^2)$. Then F is continuous and one to one (since $F'(x) = (1 + x^2) / (1 - x^2)^2 \geq 0$) and is continuous on $\mathbb{R}$. So F is a **homeomorphism** (where $\mathbb{R}$ has the standard topology and $(-1, 1)$ has the subspace topology).



Example

We give an example of a continuous bijective function which has an inverse which is **not continuous**. Consider $[0, 1)$ with the subspace topology (as a subspace of $\mathbb{R}$ with the standard topology) and $S^1 = \{(x, y) \mid x^2 + y^2 = 1\}$ with the subspace topology (as a subspace of $\mathbb{R}^2$ with the standard topology).

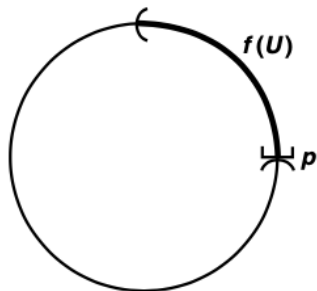
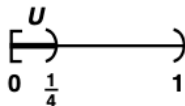
Define $f : [0, 1) \rightarrow S^1$ as $f(t) = (\cos 2\pi t, \sin 2\pi t)$.

Then f is a continuous bijection but f^{-1} is not continuous since

$$f([0, 1/2)) = \{(\cos 2\pi t, \sin 2\pi t) \mid t \in [0, 1/2)\}$$

and so the inverse image (under mapping f ; that is, $(f^{-1})^{-1}([0, 1/2)) = f([0, 1/2))$) of the open set $[0, 1/2)$ is not open.

Taken from [?]



Rules for Constructing Continuous Functions

Theorem

Let X, Y , and Z be topological spaces.

- (a) *(Constant Function)* If $f : X \rightarrow Y$ maps all of X into a single point $y_0 \in Y$, then f is continuous.
- (b) *(Inclusion)* if A is a subspace of X , the inclusion function $j : A \rightarrow X$ is continuous.
- (c) *(Composites)* If $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ are continuous, then the map $g \circ f : X \rightarrow Z$ is continuous.
- (d) *(Restricting the Domain)* If $f : X \rightarrow Y$ is continuous and if A is a subspace of X , then the restricted function $f|_A : A \rightarrow Y$ is continuous.

Note. The following result, which is fairly easy to prove, will be extremely useful in the future.

Theorem (Continued)

- (e) **(Restricting or Expanding the Range)** let $f : X \rightarrow Y$ be continuous. If Z is a subspace of Y containing the image set $f(X)$, then the function $g : X \rightarrow Z$ obtained by restricting the range of f is continuous.
- If Z is a space having Y as a subspace, then the functions $h : X \rightarrow Z$ obtained by expanding the range of f is continuous.
- (f) **(Local Formulation of Continuity)** The map $f : X \rightarrow Y$ is continuous if X can be written as the union of open sets U_α such that $f|_{U_\alpha}$ is continuous for each α .
- (g) **(The Pasting Lemma)** Let $X = A \cup B$, where A and B are closed in X . Let $f : A \rightarrow Y$ and $g : B \rightarrow Y$ be continuous. If $f(x) = g(x)$ for every $x \in A \cap B$, then f and g combine to give a continuous function $h : X \rightarrow Y$, defined by setting $h(x) = f(x)$ if $x \in A$, and $h(x) = g(x)$ if $x \in B$. This result also holds if A and B are open sets in X .

Maps into products

Theorem

Let $f : A \rightarrow X \times Y$ be given by the equation $f(a) = (f_1(a), f_2(a))$ where $f_1 : A \rightarrow X$ and $f_2 : A \rightarrow Y$. Then f is continuous if and only if the functions f_1 and f_2 are continuous.

Definition

The functions $f_1 : A \rightarrow X$ and $f_2 : A \rightarrow Y$, where $f_1 = \pi_1 \circ f$ and $f_2 = \pi_2 \circ f$ for $f : A \rightarrow X \times Y$ above, are called **coordinate functions** of f .

Proof of the theorem

Let $\pi_1 : X \times Y \rightarrow X$ and $\pi_2 : X \times Y \rightarrow Y$. Then for U open in X and V open in Y , we have $\pi_1^{-1}(U) = U \times Y$ and $\pi_2^{-1}(V) = X \times V$ open in $X \times Y$ (by the definition of product topology; these are basis elements for the product topology on $X \times Y$). So π_1 and π_2 are continuous. Note that for each $a \in A$, $\pi_1(f(a)) = \pi_1((f_1(a), f_2(a))) = f_1(a)$ and $\pi_2(f(a)) = \pi_2((f_1(a), f_2(a))) = f_2(a)$. So $f_1 = \pi_1 \circ f$ and $f_2 = \pi_2 \circ f$. Suppose f is continuous. Then, by Theorem part (c) slide 15, f_1 and f_2 are continuous.

Conversely, suppose f_1 and f_2 are continuous. Let $U \times V$ be a basis element for the product topology of $X \times Y$ (so U is open in X and V is open in Y). Now $a \in f^{-1}(U \times V)$ if and only if $f(a) \in U \times V$, or if and only if $f_1(a) \in U$ and $f_2(a) \in V$, or if and only if $a \in f_1^{-1}(U) \cap f_2^{-1}(V)$. That is, $f^{-1}(U \times V) = f_1^{-1}(U) \cap f_2^{-1}(V)$. Since f_1 and f_2 are continuous then $f_1^{-1}(U)$ and $f_2^{-1}(V)$ are open in X and so $f^{-1}(U \times V)$ is open in X . By Lemma (1) on slide 6, f is continuous.

Remark (Continuous maps on Cartesian products)

There is no useful criterion for the continuity of a map $f : A \times B \rightarrow X$ whose domain is a product space. For example, we might expect that if $f : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ is continuous if $f(x, y_0)$ and $f(x_0, y)$ are continuous for each $y_0 \in \mathbb{R}$ and $x_0 \in \mathbb{R}$ (that is, if f is continuous "in each variable separately") then $f(x, y)$ is continuous.

This is not the case!

Example:

$$f(x, y) = \begin{cases} xy / (x^2 + y^2) & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

This claim is justified in the tutorial.

Outline

- 1 Continuous functions
- 2 The metric topology
- 3 Continuity in metric spaces

The metric topology

The topological concepts you encounter in the calculus and analysis classes are based on the **metric** on $\mathbb{R}$ which gives the **distance** between x and y in $\mathbb{R}$ as $|x - y|$. **More generally**, any space with a metric on it **can have a topology defined in terms of the metric**. In this section **we define a metric and use it to create the “metric topology” on a set**.

Definition

A **metric** on a set X is a function $d : X \times X \rightarrow \mathbb{R}$ having the following properties:

- (1)** $d(x, y) \geq 0$ for all $x, y \in X$ and $d(x, y) = 0$ if and only if $x = y$.
- (2)** $d(x, y) = d(y, x)$ for all $x, y \in X$.
- (3) (The Triangle Inequality)** $d(x, y) + d(y, z) \geq d(x, z)$ for all $x, y, z \in X$.

- The nonnegative real number $d(x, y)$ is the distance between x and y .
- A set X together with a metric d form a metric space.
- Define the set $B_d(x, \varepsilon) = \{y \mid d(x, y) < \varepsilon\}$, called the ε -ball centered at x (often simply denoted $B(x, \varepsilon)$).

Definition

If d is a metric on X then the collection of all ε -balls $B_d(x, \varepsilon)$ for $x \in X$ and $\varepsilon > 0$ is a basis for a topology on X , called the metric topology induced by d .

Lemma

Let $B_d(x, \varepsilon)$ be a ε -ball in a topological space with the metric topology and metric d . Let $y \in B_d(x, \varepsilon)$. Then *there is $\delta > 0$ such that $B_d(y, \delta) \subset B_d(x, \varepsilon)$.*

Proof

Define $\delta = \varepsilon - d(x, y)$ (> 0 since $y \in B_d(x, \varepsilon)$). Then for $z \in B_d(y, \delta)$ we have, by the Triangle Inequality:

$$d(x, z) \leq d(x, y) + d(y, z) < d(x, y) + \delta = d(x, y) + \varepsilon - d(x, y) = \varepsilon.$$

So $y \in B_d(x, \varepsilon)$ and $B_d(y, \delta) \subset B_d(x, \varepsilon)$. ■

We must verify that the collection of sets in the previous definition **really satisfies the definition of basis of a topology**. First, every element $x \in X$ is in a basis element, say $B(x, 1)$. Second, let B_1 and B_2 be two basis elements and let $y \in B_1 \cap B_2$. Then from the previous lemma, there are $\delta_1 > 0$ and $\delta_2 > 0$ with $B(y, \delta_1) \subset B_1$ and $B(y, \delta_2) \subset B_2$. With $\delta = \min \{\delta_1, \delta_2\}$ we then have $B(y, \delta) \subset B_1 \cap B_2$. Since $B(y, \delta)$ is a basis element then the second part of the definition of basis is satisfied.

The following result is in line with the definition of **open set in a metric space**.

Lemma

A set U is open in the metric topology induced by metric d if and only if for each $y \in U$ there is a $\delta > 0$ such that $B_d(y, \delta) \subset U$.

Proof

Let $y \in U$. By the definition of “topology generated by a basis”, U is open if and only if there is a basis element $B_d(x, \varepsilon)$ with $y \in B_d(x, \varepsilon) \subset U$. Now, by the previous lemma on slide 23, there is $B_d(y, \delta)$ a basis element with $y \in B_d(y, \delta) \subset B_d(x, \varepsilon) \subset U$. ■

Example

For set X , define the metric

$$d(x, y) = \begin{cases} 1 & \text{if } x \neq y \\ 0 & \text{if } x = y \end{cases}$$

(this is in fact a metric). The topology reduces the [discrete topology](#) on X .

Example (Euclidean spaces $\mathbb{R}^n$)

Definition. Given $\mathbf{x} = (x_1, x_2, \dots, x_n) \in \mathbb{R}^n$, define the **Euclidean norm** of $\mathbf{x}$ as

$$\|\mathbf{x}\| = (x_1^2 + x_2^2 + \dots + x_n^2)^{1/2}.$$

Define the **Euclidean metric** on $\mathbb{R}^n$ as

$$d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\| = \left((x_1 - y_1)^2 + (x_2 - y_2)^2 + \dots + (x_n - y_n)^2 \right)^{1/2}.$$

Define the **square metric** ρ as

$$\rho(\mathbf{x}, \mathbf{y}) = \max \{ |x_1 - y_1|, |x_2 - y_2|, \dots, |x_n - y_n| \}$$

We prove that the Euclidean metric actually is a metric in the tutorials. The square metric clearly satisfies the first two parts of the definition of metric. For the Triangle Inequality, notice that for any $x_i, y_i, z_i \in \mathbb{R}$ we have by the Triangle Inequality on $\mathbb{R}$, $|x_i - z_i| \leq |x_i - y_i| + |y_i - z_i|$. So for any $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{R}^n$ we have $|x_i - z_i| \leq \rho(\mathbf{x}, \mathbf{y}) + \rho(\mathbf{y}, \mathbf{z})$ for $i = 1, 2, \dots, n$. Hence

$$\rho(\mathbf{x}, \mathbf{z}) = \max_{1 \leq i \leq n} \{|x_i - z_i|\} \leq \rho(\mathbf{x}, \mathbf{y}) + \rho(\mathbf{y}, \mathbf{z}),$$

and the Triangle Inequality holds for ρ . Therefore, ρ is a metric on $\mathbb{R}^n$.

Example (The complex plane $\mathbb{C}$)

The complex plane $\mathbb{C}$ can be identified as a set with the Euclidean plane $\mathbb{R}^2$. If z is a complex number then there exist unique real numbers a and b such that $z = a + ib$. a and b are called the real part and the imaginary part of z and are denoted by $\operatorname{Re}(z)$ and $\operatorname{Im}(z)$. Two complex numbers are said to be equal if their real and imaginary parts are equal:

$$a + ib = c + id \Leftrightarrow a = c \text{ and } b = d$$

We add (or subtract) two complex numbers by adding (or subtracting) their real and imaginary parts, and we multiply them by multiplying them out as in elementary algebra and replacing i^2 by -1 wherever it appears:

$$(a + ib) \pm (c + id) = (a \pm c) + i(b \pm d)$$

and

$$(a + ib)(c + id) = ac + iad + ibc + i^2bd = (ac - bd) + i(ad + bc).$$

Example (The complex plane $\mathbb{C}$ (Continued))

If $z = a + ib$ is a complex number, then its negative $-z$ and its conjugate $\bar{z}$ are defined by $-z = -a - ib$ and $\bar{z} = a - ib$. Simple calculations show directly that

$$\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2, \quad \overline{z_1 z_2} = \bar{z}_1 \cdot \bar{z}_2, \quad \text{and} \quad \bar{\bar{z}} = z.$$

The origin, or zero, is the complex number $0 = 0 + i0$. The ordinary distance from $z = a + ib$ to the origin is defined by

$$|z| = (a^2 + b^2)^{1/2}.$$

$|z|$ is called the absolute value or modulus of z , and it is easy to see that $|\bar{z}| = |z|$ and $|z|^2 = z\bar{z}$. The usual metric on $\mathbb{C}$ is defined by

$$d(z_1, z_2) = |z_1 - z_2|.$$

Example (The complex plane $\mathbb{C}$ (Continued))

d is a metric as a consequence of the following properties of the real function $|z|$:

- (i) $|z| \geq 0$ and $|z| = 0 \Leftrightarrow z = 0$.
- (ii) $|-z| = |z|$.
- (iii) $|z_1 + z_2| \leq |z_1| + |z_2|$.

These properties will motivate later on the introduction of a more general class of spaces that we shall call **normed** and **Banach** spaces.

Exercise (Metrics in subspaces)

Let (X, d) be a metric space and let $Y \subset X$ be a subset. Prove that

- (i) The map $d_Y = d|_{Y \times Y} : Y \times Y \longrightarrow \mathbb{R}$ obtained by restricting the metric d to Y is a metric on Y .
- (ii) The metric topology generated by d_Y on Y coincides with the subspace topology induced by the metric topology of (X, d) on Y .

Hint: Prove first that $\forall y \in Y$ and $\forall r > 0$, $B_{d_Y}(y, r) = B_d(y, r) \cap Y$.

Definition

Let X be a topological space. X is said to be **metrizable** if there exists a **metric d on a set X that induces the topology of X** . A metric space is a metrizable space X with a specific metric d that gives the topology of X .

Metrizability is a problem of **fundamental importance in topology**. One of our goals will be to find conditions for that; they are expressed there in the famous theorem called **Urysohn's metrization theorem**. Although the metrizability problem is an important problem in topology, the study of metric spaces as such belongs rather to analysis. Metrizability of a space depends only on the topology but properties that involve a specific metric for X in general do not. For instance, one can make the following definition in a metric space:

Definition

Let X be a metric space with metric d . A subset A of X is **bounded** if there is some number M such that $d(a_1, a_2) \leq M$ for every pair $a_1, a_2 \in A$. If A is bounded and nonempty, the **diameter** of A is

$$\text{diam}(A) = \sup \{d(a_1, a_2) \mid a_1, a_2 \in A\}.$$

The next result gives a metric for which every set is bounded.

Theorem

Let X be a metric space with metric d . Define $\bar{d} : X \times X \rightarrow \mathbb{R}$ by $\bar{d}(x, y) = \min\{d(x, y), 1\}$. Then $\bar{d}$ is a metric that induces the same topology as d .

Proof of the theorem

It is easy to see the first two parts of the definition of metric are satisfied by $\bar{d}$. For the third part, we need to confirm the Triangle Inequality:

$$\bar{d}(x, z) \leq \bar{d}(x, y) + \bar{d}(y, z).$$

We consider two cases.

Case 1 . If either $d(x, y) \geq 1$ or $d(y, z) \geq 1$ then the right side of this inequality is at least 1 and so the inequality holds.

Case 2 . If both $d(x, y) < 1$ and $d(y, z) < 1$ then

$$\begin{aligned} d(x, z) &\leq d(x, y) + d(y, z) \text{ by the Triangle Inequality for } d \\ &= \bar{d}(x, y) + \bar{d}(y, z). \end{aligned}$$

Since $\bar{d}(x, z) < d(x, z)$, we have the Triangle Inequality for $\bar{d}$. So $\bar{d}$ is in fact a metric.

Proof of the theorem (continued)

Let $\mathcal{B}$ be the basis for the topology induced by d and let $\mathcal{B}'$ be the basis for the topology induced by $\bar{d}$. By a preceding Lemma (Chap. 2, slide 15), the topology generated by a basis consists of all unions of basis elements. Notice that $\mathcal{B}'$ consists of all ε -balls where $\varepsilon \geq 1$, so $\mathcal{B}' \subset \mathcal{B}$ and the topology generated by $\mathcal{B}'$ is a subset of the topology generated by $\mathcal{B}$.

But for any $B_d(x, \varepsilon) \in \mathcal{B}$ we know that $B_d(x, \varepsilon)$ can be written as a union over all elements of $B_d(x, \varepsilon)$ of balls centered at these elements of $B_d(x, \varepsilon)$ of balls centered at these elements with radius less than 1 (here we use the Lemma on slide 25):

$$B_d(x, \varepsilon) = \bigcup_{y \in B_d(x, \varepsilon)} B_d(y, \delta_y)$$

where $\delta_y = \min\{\delta, 1\}$ and $B_d(y, \delta) \subset B_d(x, \varepsilon)$ as in Lemma on slide 25. So every set in the topology generated by $\mathcal{B}$ is also in the topology generated by $\mathcal{B}'$ (since $B_d(y, \delta_y) \in \mathcal{B}'$ since $\delta_y \leq 1$). So the topologies are the same and d and $\bar{d}$ induce the same topology on X . ■

The topologies on $\mathbb{R}^n$ induced by the Euclidean metric and the square metric are the same as the product topology on $\mathbb{R}^n$. To establish this we need a preliminary result.

Lemma

Let d and d' be two metrics on the set X . Let $\mathcal{T}$ and $\mathcal{T}'$ be the topologies they induce, respectively. Then $\mathcal{T}'$ is finer than $\mathcal{T}$ ($\mathcal{T}' \supset \mathcal{T}$) if and only if for each $x \in X$ and each $\varepsilon > 0$, there exists a $\delta > 0$ such that $B_{d'}(x, \delta) \subset B_d(x, \varepsilon)$.

Proof of the lemma

Suppose that $\mathcal{T}'$ is finer than $\mathcal{T}$. Let $B_d(x, \varepsilon)$ be a basis element for the metric topology $\mathcal{T}$. By a preceding Lemma (Chap. 2, slide 19) (the $(1) \Rightarrow (2)$ part) there is a basis element such that $x \in B' \subset B_d(x, \varepsilon)$. By the lemma in page 25, there is $B_{d'}(x, \delta) \subset B' \subset B_d(x, \varepsilon)$ and the first claim holds.

Suppose the ε - δ condition holds. Given a basis element B for the metric topology for $\mathcal{T}$ containing x , by the lemma in page 25 there is a basis element $B_d(x, \varepsilon) \subset B$. By the hypothesized ε - δ condition there is $B' = B_{d'}(x, \delta) \subset B_d(x, \varepsilon) \subset B$. By the Lemma (Chap. 2, slide 19) (the $(2) \Rightarrow (1)$ part), $\mathcal{T}'$ is finer than $\mathcal{T}$. ■

Theorem

The topologies on $\mathbb{R}^n$ induced by the Euclidean metric d and the square metric ρ are the same as the product topology on $\mathbb{R}^n$.

Closed balls

Definition

Let d be a metric on the set X and let $\epsilon \geq 0$. The **closed ball** $B_d[x_0, \epsilon]$ with center x_0 and radius r is the subset of X defined by

$$B_d[x_0, \epsilon] = \{x \in X \mid d(x, x_0) \leq r\}$$

$B_d[x_0, \epsilon]$ contains its center and when $r = 0$ it contains only its center. The closed balls on the real line are precisely the closed intervals. The following theorem justifies the adjective in the phrase “**closed** ball.”

Theorem

*In any metric space X , **each closed ball is a closed set.***

Proof

Let $B_d[x_0, \epsilon]$ be a closed ball in X . It suffices to show that its complement $X \setminus B_d[x_0, \epsilon]$ is open. $X \setminus B_d[x_0, \epsilon]$ is open if it is empty, so we may assume that it is non-empty.

Let x be a point in $X \setminus B_d[x_0, \epsilon]$. Since $d(x, x_0) > \epsilon$, $\epsilon_1 = d(x, x_0) - \epsilon$ is a positive real number. We take ϵ_1 as the radius of an open ball $B_d(x, \epsilon_1)$ centered on x , and we show that $X \setminus B_d[x_0, \epsilon]$ is open by showing that $B_d(x, \epsilon_1) \subseteq X \setminus B_d[x_0, \epsilon]$. Let y be a point in $B_d(x, \epsilon_1)$, so that $d(y, x) < \epsilon_1$. On the basis of this and the fact that $d(x_0, x) \leq d(x_0, y) + d(y, x)$, we see that

$$\begin{aligned} d(y, x_0) &\geq d(x, x_0) - d(y, x) > d(x, x_0) - \epsilon_1 \\ &= d(x, x_0) - [d(x, x_0) - \epsilon] = \epsilon, \end{aligned}$$

so that y is in $X \setminus B_d[x_0, \epsilon]$. ■

Important observation

One is inclined to believe that the closure $\overline{B_d(x_0, \epsilon)}$ of the open ball $B_d(x_0, \epsilon)$ is the closed ball $B_d[x_0, \epsilon]$. This **is not necessarily true**. Take the metric defined in the previous example

$$d(x, y) = \begin{cases} 1 & \text{if } x \neq y \\ 0 & \text{if } x = y \end{cases}$$

In this case and for any $x_0 \in X$, $B_d(x_0, 1) = \{x_0\} = \overline{B_d(x_0, 1)}$, while $B_d[x_0, 1] = X$. This statement **holds nevertheless true** for the structure that we introduce in the next section, that is, **normed spaces**.

Metrics induced by norms

The examples provided above with $\mathbb{R}^n$ and $\mathbb{C}$ are particular cases of **normed linear spaces**.

Definition

A **normed linear space** $(V, \|\cdot\|)$ is a vector space V on which there is defined a **norm**, that is, a function $\|\cdot\| : V \longrightarrow [0, \infty)$ which assigns to each element x in the space a real number $\|x\|$ in such a manner that

- (1) $\|x\| \geq 0$, and $\|x\| = 0 \Leftrightarrow x = 0$;
- (2) $\|x + y\| \leq \|x\| + \|y\|$;
- (3) $\|\alpha x\| = |\alpha| \|x\|$.

A normed linear space is simply a linear space in which there is available a notion of the distance from an arbitrary element to the origin.

Lemma

A *normed linear space* is a metric space with respect to the induced metric defined by

$$d(x, y) = \|x - y\|.$$

An important particular case of normed spaces are the **Banach spaces** that we introduce later on. A Banach space is a normed linear space with an additional **completeness** condition that we define later.

Example

The **Euclidean metric** in $\mathbb{R}^n$ is induced by the **Euclidean norm**. The **square metric** in $\mathbb{R}^n$ is induced by the **infinity** or the **maximum norm** $\|\cdot\|_\infty$ defined by

$$\|x\|_\infty = \max_{1 \leq i \leq n} \{|x_i|\}.$$

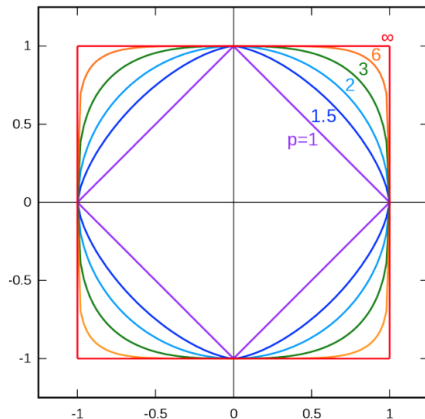
The **usual metric** on $\mathbb{C}$ is induced by the **modulus function** $|\cdot| : \mathbb{C} \rightarrow [0, \infty)$, which is a **norm** in itself.

Example (p -norms in $\mathbb{R}^n$)

The previous example is a particular case of a more general family of norms that can be introduced in $\mathbb{R}^n$. Indeed, for any $p \geq 1$, the p -norm (also called ℓ_p -norm) of vector $\mathbf{x} = (x_1, \dots, x_n)$ is defined by

$$\|\mathbf{x}\|_p = \left(\sum_{i=1}^n |x_i|^p \right)^{1/p}$$

For $p = 1$, we get the so-called **taxicab norm**, for $p = 2$, we get the **Euclidean norm**, and as p approaches ∞ the balls of p -norm approach those of the **infinity** or **maximum** norm.



These pictures show that all these norms induce the same topologies in $\mathbb{R}^2$.

A first infinite-dimensional example

Example (Sequence spaces)

Denote by $\mathbf{x} = (x_1, x_2, \dots) : \mathbb{N} \longrightarrow \mathbb{R}$ a sequence with real entries. The space $\mathbb{R}^{\mathbb{N}}$ of such sequences forms an infinite dimensional vector space with the operations:

$$\begin{aligned}\mathbf{x} + \mathbf{y} &= (x_1 + y_1, x_2 + y_2, \dots) \\ \lambda \cdot \mathbf{x} &= (\lambda x_1, \lambda x_2, \dots).\end{aligned}$$

For any $1 \leq p < \infty$, define the norm $\|\cdot\|_p : \mathbb{R}^{\mathbb{N}} \longrightarrow [0, \infty)$ by

$$\|\mathbf{x}\|_p = \left(\sum_{n \in \mathbb{N}} |x_n|^p \right)^{1/p}.$$

Example (Sequence spaces (continued))

Define by ℓ^p as the vector subspace of $\mathbb{R}^{\mathbb{N}}$ determined by the condition $\|\mathbf{x}\|_p < \infty$, that is:

$$\ell^p = \{\mathbf{x} \in \mathbb{R}^{\mathbb{N}} \mid \|\mathbf{x}\|_p < \infty\}$$

Analogously, define

$$\|\mathbf{x}\|_{\infty} = \sup_{n \in \mathbb{N}} \{|x_n|\} \quad \text{and} \quad \ell^{\infty} = \{\mathbf{x} \in \mathbb{R}^{\mathbb{N}} \mid \|\mathbf{x}\|_{\infty} < \infty\}$$

For any $1 \leq p \leq \infty$, the vector subspaces ℓ^p are normed spaces. ℓ^{∞} contains three important closed vector subspaces, namely:

- $\mathbf{c}$ is the subspace of **convergent sequences**.
- $\mathbf{c}_0$ is the subspace of **convergent sequences with zero limit**.

Outline

- 1 Continuous functions
- 2 The metric topology
- 3 Continuity in metric spaces

Continuity in metric spaces

In this section we give a number of results for metric spaces which are familiar from calculus and real analysis.

The following theorem shows that the usual ε - δ definition of continuity is equivalent to our definition of continuity in terms of inverse images of open sets when we have a metric.

Theorem

Let $f : X \rightarrow Y$. let X and Y be metrizable with metrics d_X and d_Y , respectively. Then continuity of f is equivalent to the requirement that given $x \in X$ and given $\varepsilon > 0$, there exists $\delta > 0$ such that

$$d_X(x, y) < \delta \Rightarrow d_Y(f(x), f(y)) < \varepsilon$$

Proof

Suppose f is continuous. Let $x \in X$ and $\varepsilon > 0$ be given. Consider the set $f^{-1}(B(f(x), \varepsilon))$. Since f is continuous then by definition of continuity, $f^{-1}(B(f(x), \varepsilon))$ is open in X since $B(f(x), \varepsilon)$ is open in Y . Since $x \in f^{-1}(B(f(x), \varepsilon))$ then by the lemma in page 23 there is $\delta > 0$ such that $B(x, \delta) \subset f^{-1}(B(f(x), \varepsilon))$. Then $d_X(x, y) < \delta$ implies $y \in B(x, \delta)$, so $f(y) \in B(f(x), \varepsilon)$ and $d_Y(f(x), f(y)) < \varepsilon$.

Conversely, suppose that the ε - δ condition is satisfied. Let V be an open set in Y . Let us show that $f^{-1}(V)$ is open. Given $x \in f^{-1}(V)$, we have $f(x) \in V$. Since V is open and $f(x) \in V$ then by the lemma in page 25 there is $\varepsilon > 0$ such that $B(f(x), \varepsilon) \subset V$. By the ε/δ hypothesis, there is $\delta > 0$ such that $d_X(x, y) < \delta$ implies $d_Y(f(x), f(y)) < \varepsilon$ (i.e., $f(y) \in B(f(x), \varepsilon)$). In other words: $f(B(x, \delta)) \subset B(f(x), \varepsilon)$. So $x \in B(x, \delta) \subset f^{-1}(B(f(x), \varepsilon)) \subset V$. Therefore, by the lemma on slide 25, $f^{-1}(V)$ is open and so f is continuous. ■

Bounded maps

When the range of a map is a metric space we can talk about **boundedness of that map**.

Definition

Let X be a set and let (Y, d) be a metric space. We say that a function $f : X \rightarrow Y$ is **bounded** if its image $f(X)$ is a **bounded subset of the metric space (Y, d)** . The **set of bounded functions** between X and (Y, d) is denoted by $\mathcal{B}(X, Y)$.

Example

The function $\sin : \mathbb{R} \rightarrow \mathbb{R}$ is bounded while $\exp : \mathbb{R} \rightarrow \mathbb{R}$ is not when $\mathbb{R}$ is endowed with the standard metric.

Example (Another infinite dimensional normed space)

Let X be a set and let $\mathbb{R}$ considered as a normed space with the absolute value. The set $\mathcal{B}(X, \mathbb{R})$ forms an infinite-dimensional vector space with the operations defined by:

$$\begin{aligned}(f + g)(x) &= f(x) + g(x), \\ (\lambda f)(x) &= \lambda \cdot f(x).\end{aligned}$$

The map $\|\cdot\|_{\infty} : \mathcal{B}(X, \mathbb{R}) \longrightarrow [0, \infty)$ defined by

$$\|f\|_{\infty} = \sup_{x \in X} \{|f(x)|\}$$

makes it into a normed space. (See [?, Exercise 9.5])

Convergence in metric spaces: The Sequence Lemma

Recall from real and complex analysis that a **closed set contains its limit points**. The following result for topological spaces is related to this familiar property from $\mathbb{R}$.

Lemma (The Sequence Lemma)

Let X be a topological space. Let $A \subset X$. If there is a sequence of points of A converging to x , then $x \in \bar{A}$. If X is metrizable and $x \in \bar{A}$ then there is a sequence $\{x_n\} \subset A$ such that $\{x_n\} \rightarrow x$.